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Dipole hyperfine coupling tensor

intro

We can use classical magnetostatics to calculate the magnetic induction $\mathbf{B}$ at the nuclear site due to a distant magnetic dipole $\mathbf{m}$ (due to a electron spin/magnetic moment). We will then pull out the magnetic moment, and define the dipolar hyperfine coupling tensor, which will be composed of matrix elements that include everything but the magnetic moment vector.

We start with the vector potential of a classical magnetic dipole moment located at the origin in the far-field, as derived in equation 5.55 of Jackson (Jackson_1998_classical_electrodynamics_3rd):

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3} \quad (1)$$

where μ_0 is the permeability of free space, $\mathbf{r}$ is the position vector (and $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$). Taking the curl of the vector potential gives us the magnetic induction:

$$\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A}(\mathbf{r}) \quad (2)$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \nabla \times \left(\frac{\mathbf{m} \times \mathbf{r}}{r^3} \right) \quad (3)$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \left(\frac{3\mathbf{r}(\mathbf{m} \cdot \mathbf{r})}{r^5} - \frac{\mathbf{m}}{r^3} \right) \quad (4)$$

Now need to choose how to proceed, vector identities or matrix multiplication; considering that we would like to end up with a matrix, I will take that approach. We can write the cross products in terms of matrices (Liu_2008_HadamardKhatriRao) as follows:

$$\mathbf{a} \times \mathbf{b} = [\mathbf{a}]_{\times} \mathbf{b} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} \quad (5)$$

where we have defined the cross product matrix as:

$$[\mathbf{a}]_{\times} \equiv \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \quad (6)$$

So first we use the anticommutative property of the cross product $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ to rewrite equation 3 as

$$\mathbf{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \nabla \times \left(\frac{\mathbf{r} \times \mathbf{m}}{r^3} \right) \quad (7)$$

Then we can replace the cross products as matrices:

$$\mathbf{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \left(\begin{bmatrix} 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} \\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{bmatrix} \left(\begin{bmatrix} 0 & -\frac{z}{r^3} & \frac{y}{r^3} \\ \frac{z}{r^3} & 0 & -\frac{x}{r^3} \\ -\frac{y}{r^3} & \frac{x}{r^3} & 0 \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} \right) \right) \quad (8)$$

Using the associative property of matrix multiplication, we can first multiply the two (3, 3) matrices together and pull out the magnetic moment vector---a (3, 1) matrix:

$$\mathbf{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \left(\left(\begin{bmatrix} 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} \\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{bmatrix} \begin{bmatrix} 0 & -\frac{z}{r^3} & \frac{y}{r^3} \\ \frac{z}{r^3} & 0 & -\frac{x}{r^3} \\ -\frac{y}{r^3} & \frac{x}{r^3} & 0 \end{bmatrix} \right) \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} \right) \quad (9)$$

Now we have the proper form for the magnetic induction $\mathbf{B}(\mathbf{r}) = \mathbb{A}_{\text{dip}} \cdot \mathbf{m}$ and all that is left to do is to perform the matrix multiplication and evaluate the partial derivatives to extract $\mathbb{A}_{\text{dip}}$:

$$\mathbb{A}_{\text{dip}} = -\frac{\mu_0}{4\pi} \begin{bmatrix} 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} \\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{bmatrix} \begin{bmatrix} 0 & -\frac{z}{r^3} & \frac{y}{r^3} \\ \frac{z}{r^3} & 0 & -\frac{x}{r^3} \\ -\frac{y}{r^3} & \frac{x}{r^3} & 0 \end{bmatrix} \quad (10)$$

$$= -\frac{\mu_0}{4\pi} \begin{bmatrix} 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} \\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{bmatrix} \begin{bmatrix} 0 & -z(x^2 + y^2 + z^2)^{-\frac{3}{2}} & y(x^2 + y^2 + z^2)^{-\frac{3}{2}} \\ z(x^2 + y^2 + z^2)^{-\frac{3}{2}} & 0 & -x(x^2 + y^2 + z^2)^{-\frac{3}{2}} \\ -y(x^2 + y^2 + z^2)^{-\frac{3}{2}} & x(x^2 + y^2 + z^2)^{-\frac{3}{2}} & 0 \end{bmatrix} \quad (11)$$

$$\mathbb{A}_{\text{dip}}^{xx} = -\frac{\mu_0}{4\pi} \left(0 - \frac{\partial}{\partial z} \left(z(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) + \frac{\partial}{\partial y} \left(-y(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) \right) \quad (12)$$

$$\mathbb{A}_{\text{dip}}^{xy} = -\frac{\mu_0}{4\pi} \left(0 - \frac{\partial}{\partial z} (0) + \frac{\partial}{\partial y} \left(x(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) \right) \quad (13)$$

$$\mathbb{A}_{\text{dip}}^{xz} = -\frac{\mu_0}{4\pi} \left(0 - \frac{\partial}{\partial z} \left(-x(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) + \frac{\partial}{\partial y} (0) \right) \quad (14)$$

$$\mathbb{A}_{\text{dip}}^{yx} = -\frac{\mu_0}{4\pi} \left(\frac{\partial}{\partial z} (0) + 0 - \frac{\partial}{\partial x} \left(-y(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) \right) \quad (15)$$

$$\mathbb{A}_{\text{dip}}^{yy} = -\frac{\mu_0}{4\pi} \left(\frac{\partial}{\partial z} \left(-z(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) + 0 - \frac{\partial}{\partial x} \left(x(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) \right) \quad (16)$$

$$\mathbb{A}_{\text{dip}}^{yz} = -\frac{\mu_0}{4\pi} \left(\frac{\partial}{\partial z} \left(y(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) + 0 - \frac{\partial}{\partial x} (0) \right) \quad (17)$$

$$\mathbb{A}_{\text{dip}}^{zx} = -\frac{\mu_0}{4\pi} \left(-\frac{\partial}{\partial y} (0) + \frac{\partial}{\partial x} \left(z(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right) + 0 \right) \quad (18)$$

imports

```
In [1]: import sympy as sym
import numpy as np
```

calculate the partial derivatives

define symbols

```
In [2]: x = sym.Symbol(r'x', real=True)
y = sym.Symbol(r'y', real=True)
z = sym.Symbol(r'z', real=True)

mu0 = sym.Symbol(r'\mu_0', real=True)
```

get matrix

```
In [3]: # include just the relevant parts without constants mu_0/4pi
A_dip_xx_raw = sym.simplify(0 - sym.diff(z*(x**2 + y**2 + z**2)**sym.Rational(-3/2), z) +
                                sym.diff(-y*(x**2 + y**2 + z**2)**sym.Rational(-3/2), y))
A_dip_xy_raw = sym.simplify(0 - 0 + sym.diff(x*(x**2 + y**2 + z**2)**sym.Rational(-3/2), y))
A_dip_xz_raw = sym.simplify(0 - sym.diff(-x*(x**2 + y**2 + z**2)**sym.Rational(-3/2), z) + 0)
A_dip_yx_raw = sym.simplify(0 + 0 - sym.diff(-y*(x**2 + y**2 + z**2)**sym.Rational(-3/2), x))
A_dip_yy_raw = sym.simplify(sym.diff(-z*(x**2 + y**2 + z**2)**sym.Rational(-3/2), z) + 0 -
                                sym.diff(x*(x**2 + y**2 + z**2)**sym.Rational(-3/2), x))
A_dip_yz_raw = sym.simplify(sym.diff(y*(x**2 + y**2 + z**2)**sym.Rational(-3/2), z) + 0 - 0)
A_dip_zx_raw = sym.simplify(0 + sym.diff(z*(x**2 + y**2 + z**2)**sym.Rational(-3/2), x) + 0)
A_dip_zy_raw = sym.simplify(-sym.diff(-z*(x**2 + y**2 + z**2)**sym.Rational(-3/2), y) + 0 + 0)
A_dip_zz_raw = sym.simplify(-sym.diff(y*(x**2 + y**2 + z**2)**sym.Rational(-3/2), y) +
                                sym.diff(-x*(x**2 + y**2 + z**2)**sym.Rational(-3/2), x) + 0)

# reconstruct the matrix including the prefactors
A_dip = -mu0/(4*sym.pi)*sym.Matrix([A_dip_xx_raw, A_dip_xy_raw, A_dip_xz_raw],
                                    [A_dip_yx_raw, A_dip_yy_raw, A_dip_yz_raw],
                                    [A_dip_zx_raw, A_dip_zy_raw, A_dip_zz_raw])
```

In [4]: A_dip

Out[4]:

$$\begin{bmatrix} -\frac{\mu_0(-2x^2+y^2+z^2)}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} & \frac{3\mu_0xy}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} & \frac{3\mu_0xz}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} \\ \frac{3\mu_0xy}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} & -\frac{\mu_0(x^2-2y^2+z^2)}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} & \frac{3\mu_0yz}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} \\ \frac{3\mu_0xz}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} & \frac{3\mu_0yz}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} & -\frac{\mu_0(x^2+y^2-2z^2)}{4\pi(x^2+y^2+z^2)^{\frac{5}{2}}} \end{bmatrix}$$

So each matrix element has the term $(x^2 + y^2 + z^2)^{-5/2} = r^{-5}$, and the diagonal matrix elements have terms like (for example in $\mathbb{A}_{\text{dip}}^{zz}$):

$$-x^2 + -y^2 + 2z^2 = -x^2 - y^2 - z^2 + 3z^2 = -(x^2 + y^2 + z^2) + 3z^2 = -r^2 + 3z^2 \quad (21)$$

final form of the dipole hyperfine coupling tensor for nuclear spin at position $\mathbf{r}$ due to a magnetic moment at the origin:

$$\mathbb{A}_{\text{dip}} = \frac{\mu_0}{4\pi} \begin{bmatrix} \frac{3x^2-r^2}{r^5} & \frac{3xy}{r^5} & \frac{3xz}{r^5} \\ \frac{3yx}{r^5} & \frac{3y^2-r^2}{r^5} & \frac{3yz}{r^5} \\ \frac{3zx}{r^5} & \frac{3zy}{r^5} & \frac{3z^2-r^2}{r^5} \end{bmatrix} \quad (22)$$

and the magnetic induction:

$$\mathbf{B}(\mathbf{r}) = \mathbb{A}_{\text{dip}} \cdot \mathbf{m} \quad (23)$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \begin{bmatrix} \frac{3x^2-r^2}{r^5} & \frac{3xy}{r^5} & \frac{3xz}{r^5} \\ \frac{3yx}{r^5} & \frac{3y^2-r^2}{r^5} & \frac{3yz}{r^5} \\ \frac{3zx}{r^5} & \frac{3zy}{r^5} & \frac{3z^2-r^2}{r^5} \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} \quad (24)$$

However, we have one final issue, we actually want to find the hyperfine coupling tensor the other way around, i.e. a magnetic moment at position $\mathbf{r}$ and the hyperfine coupling of said spin to the nucleus at the origin. However, this is equivalent to a translation from $\mathbf{r} \rightarrow -\mathbf{r}$. As can be seen above in equation 24, the magnetic induction $\mathbf{B}(\mathbf{r}) = \mathbf{B}(-\mathbf{r})$. Therefore, equation 24 is our answer.

Check units

Quick check of the units for our calculation for posterity. We will do the calculation in $\text{\AA}$, meaning that the final matrix elements (not including the prefactor of $\frac{\mu_0}{4\pi}$) has units of $\text{\AA}^{-3}$. So need to transform to m^{-3} , and then we would like to give the magnetic moments in Bohr magnetons μ_B , so need to account for that by multiplying by μ_B in SI units.

In [5]:

```
# angstroms
AA = sym.Symbol(r'\mathring{A}', real=True)

# SI base units
kg = sym.Symbol(r'kg', real=True)
m = sym.Symbol(r'm', real=True)
s = sym.Symbol(r's', real=True)
A = sym.Symbol(r'A', real=True)

# composite SI units
# Henry
H = kg*m**2/s**2/A**2
# Joule
J = kg*m**2/s**2
# Tesla
T = kg/A/s**2

mu0 = 1.25663706212e-6*H/m # vacuum permeability (4π × 1.00000000055(15)×10-7 H·m-1 )
muB = 9.274009994e-24*J/T # Bohr magneton in SI units

muB_symbol = sym.Symbol(r'\mu_B', real=True) # symbolic muB for units
```

In [6]:

```
# So we want to go from the raw calculation units of  $\text{\AA}^{-3}$ 
# to T/muB

mu0/(4*np.pi)*(1/AA**3)*(1e10*AA/m)**3*(muB/muB_symbol)
```

Out[6]:

$$\frac{0.927400999904855 kg}{A \mu_B s^2}$$

$\frac{kg}{As^2} = T$ So our units are now T/μ_B . So the final multiplicative factor at the second to last step in our calculation (getting the units of the

individual A_i at the individual magnetic lattice sites i is as follows:

```
In [7]: # Ai = mu0/(4*np.pi)*Ai_raw*1e10**3*mu_B
```

```
In [ ]:
```